

Image Plagiarism Detection with Equivariant Encoders

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Goals

Plagiarists modify images by rotating, reflecting, shifting, and re-compressing them. A good detector must stay invariant to these transformations.

Hypothesis

Embedding exact G -invariance ($G \in \{D_4, SE(2)\}$) into a ViT encoder reduces FPR and raises Recall on geometric transformations from G , compared to a non-equivariant ViT-Tiny/16 trained with geometric augmentation.

Problem

Standard ViTs have no geometric invariance; augmentation only covers group orbits stochastically and gives no guarantee on unseen transforms.

Solution

- 1) Build a siamese detector with encoders spanning $D_4, SE(2)$.
- 2) Train each with and without geometric augmentations.
- 3) Evaluate FPR and Recall on balanced DomainNet.

Central Idea - Equivariant Encoders

Let G act on the image space $\mathcal{X}$ and let $\rho: G \rightarrow \text{GL}(d)$ be a linear representation of G on $\mathbb{R}^d$. An encoder $\phi: \mathcal{X} \rightarrow \mathbb{R}^d$ is G -equivariant if

$$\phi(g \cdot x) = \rho(g) \phi(x) \quad \forall x \in \mathcal{X}, \forall g \in G.$$

- 1) Inputs: pair (x_1, x_2) ; label $y \in \{0, 1\}$.
- 2) Encoder: ϕ_θ , shared, G -invariant.
- 3) Head: symmetric fusion h_ψ , MLP.
- 4) Loss: weighted BCE + $\lambda \mathcal{L}_{\text{contr}}^{L^2}$.
- 5) Groups: D_4 (Octic), $SE(2)$ (Harmformer).

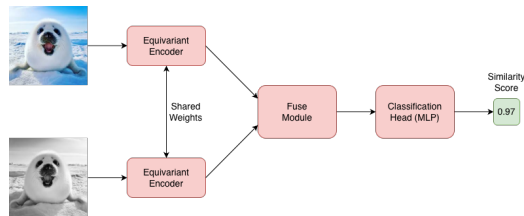


Figure: Siamese pipeline: shared equivariant encoder, fuse module, classification head $\rightarrow$ similarity score.

Related Work

- 1) **Bromley et al., 1993** — siamese networks with shared weights and learned pairwise distance.
- 2) **Dosovitskiy et al., 2021** — Vision Transformer (ViT): image patches as tokens, transformer encoder, state-of-the-art on ImageNet at scale; foundation for ViT backbones in vision.
- 3) **Nordström et al., 2025 (Octic ViT)** — D_4 -equivariant ViT; 40% FLOPs reduction.
- 4) **Karella et al., 2024 (Harmformer)** — $SE(2)$ -equivariance via circular harmonic filters.

Problem Statement

Research hypothesis. For each $G \in \{D_4, SE(2)\}$, let ϕ^G be a G -invariant ViT encoder trained with non-group augmentations only, and ϕ^{aug} a ViT-Tiny/16 trained with non-group + group- G augmentations. Then

$$\text{FPR}(\phi^G) \leq \text{FPR}(\phi^{\text{aug}}), \quad \text{Recall}(\phi^G) \geq \text{Recall}(\phi^{\text{aug}}).$$

Solution

Model. Siamese detector with shared G -invariant ϕ_θ and symmetric head h_ψ .

Let $z_i = \phi_\theta(x_i)$; build pair features

$$\delta = |z_1 - z_2|, \quad p = z_1 \odot z_2 \text{ (element-wise)}, \quad v = [u_\delta(\delta); u_p(p)] \in \mathbb{R}^{2d},$$

The *fusion* MLP is $w_\psi: \mathbb{R}^{2d} \rightarrow [0, 1]$, $w_\psi(v) = \sigma(c^\top \text{ReLU}(Av + b) + d)$

The detector output is

$$f_\theta(x_1, x_2) = w_\psi(v) = h_\psi(\phi_\theta(x_1), \phi_\theta(x_2)), \quad \phi_\theta(g \cdot x) = \phi_\theta(x) \quad \forall g \in G.$$

Loss. $\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{contr}}^{L^2}$, where $z_i = \phi_\theta(x_i)$ and

$$\mathcal{L}_{\text{contr}}^{L^2}(z_1, z_2, y) = y \|z_1 - z_2\|_2^2 + (1-y) \max(0, m - \|z_1 - z_2\|_2)^2.$$

Computational Experiment

Encoders. ViT-Tiny/16 (baseline), OcticViT (D_4), Harmformer ($SE(2)$).

Setup.

- ▶ Training set: COCO 2017 (~118K images).
- ▶ Test set: DomainNet, 6 domains, 1 200 pairs (100 pos + 100 neg per domain).
- ▶ Augmentation pipeline (each transform sampled independently, $p = 0.3$):
 - Non-group* (shared, all runs): CJ, GBlur, GN, JPEG, WM.
 - Group* (added to ViT-Tiny/16 per comparison):
 - D_4 : R90/180/270, HFlip, VFlip.
 - $SE(2)$: R90/180/270, HFlip, VFlip, Crop, Scale.

Error Analysis

Per-group comparison. For each

$G \in \{D_4, SE(2)\}$:

- 1) Train ϕ^G with non-group augs only.
- 2) Train ViT-Tiny/16 with non-group + block- G augs.
- 3) Full ROC and PR curves; report AUC.

Non-group augs (shared): CJ, GBlur, GN, JPEG, WM.

Per-class evaluation.

At test time, generate positive pairs using a single known augmentation. Report FPR and Recall separately for each transform class (heatmap: encoder $\times$ augmentation).

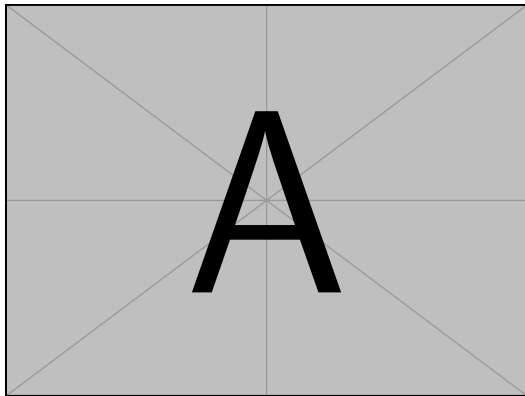


Figure: ROC curves per encoder (placeholder).



Models, dataset, criteria

Encoder	Group augs given to ViT-Tiny/16	FPR	Recall
<i>Comparison 1: $G = D_4$</i>			
ViT-Tiny/16 + D_4 augs	R90/180/270, HFlip, VFlip	—	—
OcticViT (no group augs)	—	—	—
<i>Comparison 2: $G = SE(2)$</i>			
ViT-Tiny/16 + $SE(2)$ augs	R90/180/270, HFlip, VFlip, Crop, Scale	—	—
Harmformer (no group augs)	—	—	—

All encoders share non-group augmentations (CJ, GBlur, GN, JPEG, WM).

Results and conclusions

Results. Placeholders pending training runs:

- ▶ FPR and Recall per encoder — *to be filled*.
- ▶ Per-transformation Recall heatmap (encoder $\times$ aug. class) — *to be filled*.
- ▶ ROC and PR curves per comparison (ϕ^G vs. ϕ^{aug}) — *to be filled*.

Expected conclusions.

- 1) For each G , the G -invariant encoder trained without group augs matches or exceeds the augmented ViT-Tiny/16 in both FPR and Recall.
- 2) Adding block- G augs to ϕ^G yields no gain.